
AUTOMATED DISRUPTION LABELING OF DIII-D SHOTS USING A 1D CONVOLUTIONAL NEURAL NETWORK AND TEMPORAL FILTERING OF PLASMA CURRENT

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ABSTRACT

Experimental Tokamak fusion facilities like DIII-D frequently experience disruption events that result in plasma current dropping to zero. We introduce a computationally inexpensive model for calculating disruption probability and accurately identifying time of disruption in a plasma shot. We use a labeled plasma disruption dataset from the DIII-D Tokamak reactor comprising shot numbers 125500–186000 (nearly 40 000 total plasma shots) to develop a new disruption labeling model that generates automated disruption labels using only plasma current time-series data. Our CNN (Convolutional Neural Network) model performs boolean disruption/not-disruption prediction and achieves a 99% recall and 91% precision on our test dataset with an AUC of 0.99. We additionally develop a classical convolutional filter heuristic capable of accurately identifying time of disruption with median 0 and standard deviation 1.0 ms on our test dataset. Our model can assist the scientific community by rapidly auto-labeling shots for further analysis.

1 Introduction

A Tokamak is a torus-shaped magnetic confinement fusion reactor and is a leading candidate to produce controlled thermonuclear fusion power [1]. A prominent challenge facing Tokamak reactors is the issue of disruption events caused by turbulent instability within the plasma. Such disruption events cause the internal reactor current ($I(t)$) to drop to zero and can potentially cause damage to the interior of the reactor. Abnormalities such as plasma disruptions are a prominent challenge for upcoming flagship fusion experiments such as the International Thermonuclear Experimental Reactor (ITER) [2, 1], which requires a 30 ms warning time for successful disruption mitigation [3].

Decades of Tokamak experiments have created a large corpus of fusion data, leading to a growing body of literature utilizing machine learning for problems infeasible to address with classical methods, including disruption predic-

tion [4]. A variety of classical machine learning algorithms have demonstrated promise; support vector machines [5], fuzzy logic [6], generative topographic mapping [7], and artificial neural networks have all been studied on a broad range of Tokamaks across the United States, India, China, and Europe [8]. In the deep learning space, sequence based models such as Long-Short Term Memory Networks (LSTMs) show promise for dynamic time-series forecasting of plasma disruptions [9]. A variety of approaches have also been proposed for knowledge transfer from existing Tokamaks to new ones, using hybrid architectures [10] and transfer learning [11].

Convolutional Neural Network (CNN) architectures particularly excel at disruption classification, commonly predicting disruptive shots with over 90% accuracy [12, 13, 14]. Such classifier models will assign each plasma shot with a disruption probability P_{disrupt} and flag it as disruptive if $P_{\text{disrupt}} \leq K$ given decision threshold $K \in [0, 1]$. A true positive is a disruptive plasma shot correctly classified as

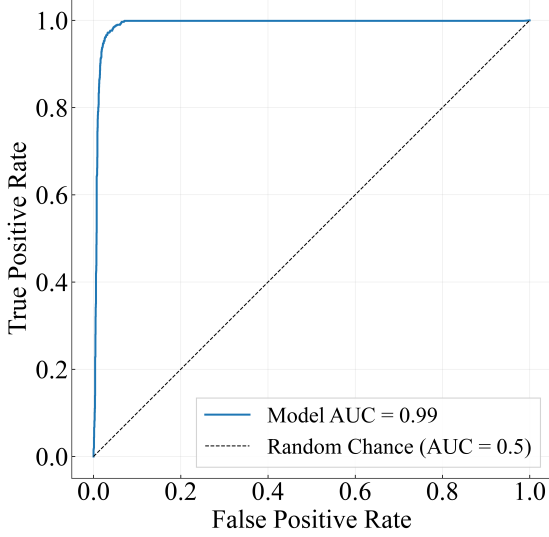


Figure 1: The ROC curve for our disruption classifier. The integral (AUC) of the ROC characterizes a classifier’s performance, with a score of 1 indicating 100% true positives and 0% false negatives. Our model has an AUC of 0.9902, close to the hypothetical maximum.

disruptive, while a false positive is a non-disruptive plasma shot incorrectly identified as disruptive; the True Positive Rate $TPR(K)$ is then the ratio of total true positives to total disruptive shots (commonly referred to as recall) while the False Positive Rate $FPR(K)$ is the ratio of total false positives to total non-disruptive shots. The ROC (Receiver-Operator) curve is a graph of $FPR(K)$ versus $TPR(K)$ for each $K \in [0, 1]$. The integral of the ROC curve is defined as the model’s AUC (Eq (1)) [15], a standard metric for characterizing classifier performance throughout machine learning literature [16]. The ideal classifier has an AUC of 1 (100% true positives and 0% false positives).

$$AUC = \int_0^1 TPR(K) d(FPR(K)) \quad (1)$$

A large fraction of the data produced by long-running Tokamak fusion facilities such as the DIII-D reactor operated by General Atomics has already been labeled for training deep learning models for disruption analysis. A team at Princeton trained a Convolutional Neural Network on Electron Cyclotron Emission imaging (ECEi) diagnostic data on 2747 DIII-D shots, achieving an AUC of 0.96 with a multi-layer long/short-term memory network (LSTM) [12]. Another team at MIT trained a neural network on 14 plasma signatures from 22 669 shots from the C-Mod, DIII-D, and EAST Tokamaks, correctly identifying 84% of disruptive events with a model AUC of 0.94 and an early warning time of 15 ms [17]. Similarly at Harvard, researchers achieved an AUC of 0.94 on time-series disruption forecasting using an LSTM trained on 8959 shots from DIII-D and JET, optimizing models for performance at deadlines of both 30 ms and 1 s [18].

Our CNN does not perform forecasting, but outperforms each aforementioned classification result with an exceptional AUC of 0.99 (Figure 1). Such high AUC scores are more typically seen in binary classification tasks where a large amount of structured, relatively homogeneous data is available, in fields such as astrophysics [19] and medicine [20]. Our dataset of nearly 40 000 plasma shots combined with our simple 1D input stream enables us to achieve similar results.

2 Neural Net Classifier

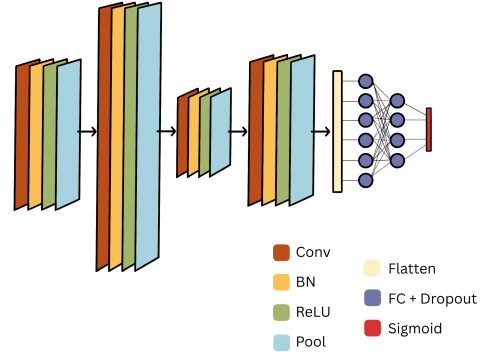


Figure 2: Model architecture for our disruption labeling Convolutional Neural Network. We have four convolutional layers and two fully connected layers, and utilize max pooling, batch normalization, and dropout, along with a ReLU activation function. The model input is normalized current sampled at 4 kHz and the output is disruption probability for the shot.

Our dataset consisted of 39 438 shot files from the DIII-D team, of which 13 429 are disruptive shots with a labeled disruption time t_D , and the rest are non-disruptive with a labeled disruption time $t_D = -1$. A 2019 paper by Kates-Harbeck et al. [21] uses the same dataset. The labels appear to be generated through a combination of manual and automated methods, as evidenced by the presence of incorrect labels in a number of shots (Figure 3).

Each shot in our dataset sampled reactor current at 4 kHz, giving us a discrete $1 \times m$ current vector $\mathbf{I}$ where the m th component is current at the m th timestep.

$$\mathbf{I}_p = [I_1 \quad \dots \quad I_m]^T \quad (2)$$

We used an 80/10/10 train/test/dev split, which left us with 31542 shots in the training set. To prepare our data for the neural net classifier, we tested three different normalization methods, including per-shot min-max unit scale normalization, per-shot Mean-Variance (meanvar) normalization, and dataset-wide meanvar normalization. We found most success with the dataset-wide meanvar approach. Letting the current mean and standard deviation across all shots

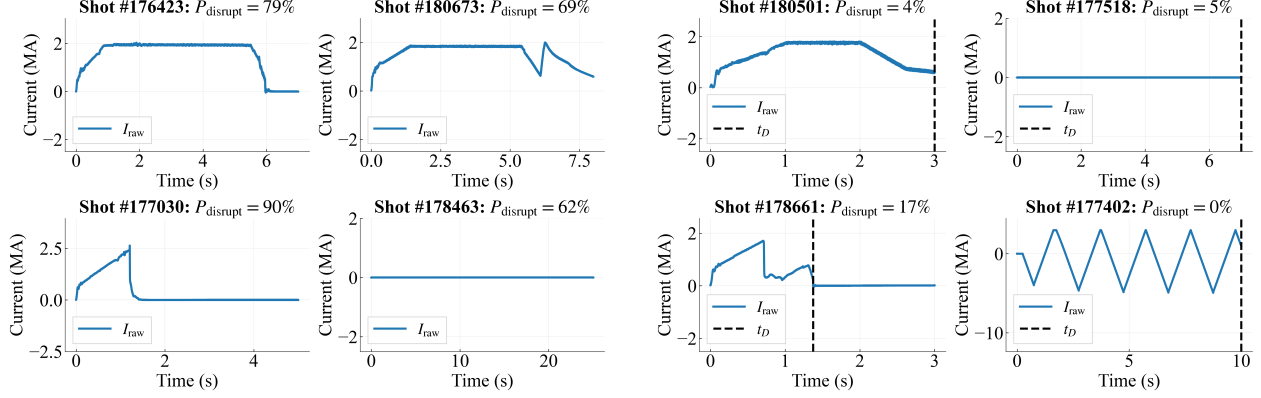


Figure 3: On the left, four selected non-disruptive shots incorrectly identified as disruptive by the CNN. Shots 176423 and 177030 feature a drop resembling a disruption, thereby confusing our model, while shot 178463 appears to be garbage data given that the current peaks at ≈ 0.02 MA (3a). On the right, four selected disruptive shots incorrectly identified as non-disruptive by the CNN. Each shot lacks the characteristic drop to zero that typically identifies a disruption, while shots 180501 and 177403 appear to be erroneously labeled and lack an evident disruption (3b).

be μ and σ , we computed a normalized shot tensor I_N for the arbitrary i th shot with N_i timesteps. If the longest shot has N timesteps, we pad the i th shot with $N - N_i$ trailing zeroes.

$$I_{p,N} = \begin{bmatrix} \frac{1}{\sigma}(I_p - \mu) \\ \mathbf{0}_{(N-N_i) \times 1} \end{bmatrix} \quad (3)$$

The normalized data was then fed into a Convolutional Neural Network (see Figure 2) consisting of four convolutional layers with stride 1, each with batch normalization [22], a Rectified Linear Unit (ReLU) activation function [23], and a final max pool layer [24]. The convolution outputs were then fed into two fully connected layers with batch normalization, ReLU, and a dropout layer with a tunable dropout rate dropout hyperparameter. The code was written in PyTorch [25] and trained using the Adam (Adaptive Moment Estimation) optimizer [26] with weight decay and a learning rate tuned per-epoch with ReduceLROnPlateau in "min" mode [27]. We utilized the BCEWithLogitsLoss loss function with configurable positive weight and gradient clipping, and stopped training when enough consecutive epochs showed no improvement.

We tuned our hyperparameters with Bayesian Optimization using the ExpectedImprovement acquisition function with $\xi = 0.05$ [28]. We additionally inserted random samples every 5 trials to escape local minima. Our scoring strategy was a weighted F_β [29], where parameter β is the weighted harmonic mean of precision and recall (Eq (4)). We chose $\beta = 1.5$ to prioritize recall (R) over precision (P) and minimize false negatives, rejecting trials below a precision floor of 90%.

$$F_\beta = \frac{PR(1 + \beta^2)}{P\beta^2 + R} \quad (4)$$

We tuned model and architecture hyperparameters for 100+ trials on the Polaris supercomputer in Argonne National Lab, with each trial sampling fresh hyperparameters and training a new model for up to 200 epochs. Increasing the decision threshold gave us higher recall at the cost of precision. With a decision threshold of 50%, our best model achieved a precision of 90.52% and recall of 98.84% on the test set, along with an F1 score of 0.9450 and F2 score of 0.9706. Due to our large dataset and relatively simple feature mapping, we see an unusually high AUC of 0.9902 (Figure 1). The tuned hyperparameters are presented in Table 1.

Table 1: Tuned Hyperparameters

Hyperparameter	Value
Learning Rate	0.0005
Number of Epochs	200
Dropout Rate	0.35
Weight Decay	0.0001
Training Batch Size	128
Gradient Clip	1.0
Learning Rate Scheduler Factor	0.5
Learning Rate Patience	5
Early Stopping Patience	10
Positive Weight	3.0

3 Convolutional Heuristic for Time of Disruption

For any given shot, let the current be $I(t)$, $t \in [0, T]$. We start by noting that shots with $\max I(t) \approx 0$ are negative polarity shots and must be flipped before proceeding (Eq (5)).

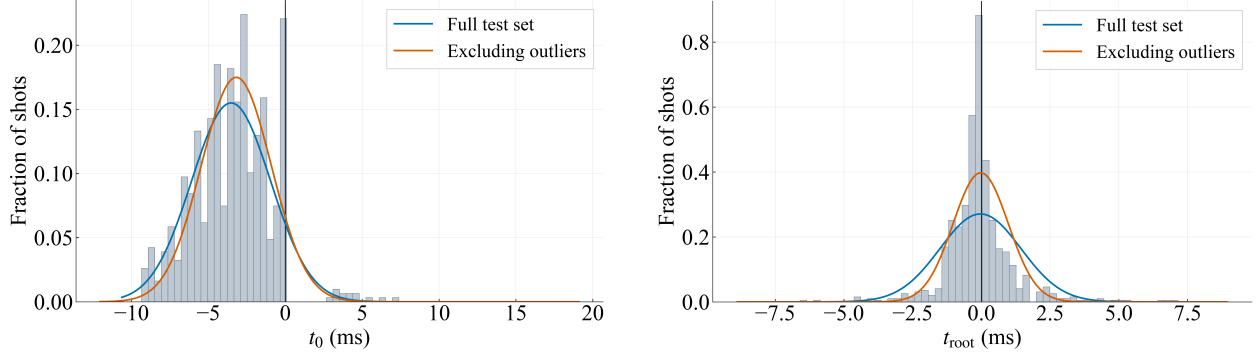


Figure 4: On the left, a histogram of the difference $\Delta t = t_0 - t_D$ (See Eq (8)). Most of our predictions are early, with a mean $\Delta t \approx -3.9$ ms (4a). On the right, a histogram of the difference $\Delta t = t_{\text{root}} - t_D$. The Gaussian is centered with $\mu \approx 0.1$ ms, $\sigma = 1.0$ ms (4b). For both charts, we plot a Gaussian fit over the entire test set (blue) and a Gaussian fit excluding outliers where $|\Delta t| > 3\sigma$ (orange).

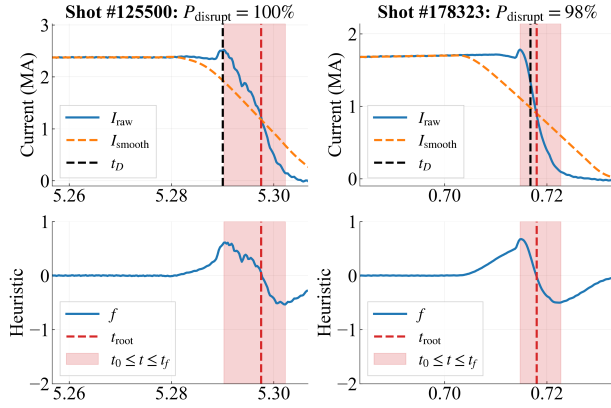


Figure 5: Two example shots (above) with their disruption heuristic $f(t)$ (below). On the left, we see shot #125500 with a DIII-D label near the disruption start (Eq (8)); on the right, we see shot #178323 with a DIII-D label near the disruption root (Eq (10)).

$$I_{\text{raw}}(t) = \begin{cases} -I(t) & \max(I) \approx 0 \\ I(t) & \text{otherwise} \end{cases} \quad (5)$$

We then calculate a smoothed current $I_{\text{smooth}}(t)$ by applying a moving average. We compute the moving average on our dataset as a discrete convolution with window size $K = T/S$, where $S = 300$ is an experimentally determined smoothing parameter. Applying `np.convolve` on the current tensor Eq (2), we get the i th component of the smooth current $\mathbf{I}_{p,\text{smooth}}$ given by Eq 6.

$$\mathbf{I}_{p,\text{smooth}_i} = \frac{1}{K} \sum_{k=-K/2}^{K/2} I_{p,i+k} \quad (6)$$

We then define the time of disruption heuristic filter $f(t)$ as the difference between the raw current in Eq (5) and smooth current in Eq (6).

$$f(t) = I_{\text{raw}}(t) - I_{\text{smooth}}(t) \quad (7)$$

The global maximum of $f(t)$ is our predicted disruption start $t = t_0$, and the first local minimum where $t > t_0$ is our predicted disruption end t_f .

$$t_0 = \max(f(t)) \quad (8)$$

$$t_f = \min(f(t)), \quad t > t_0 \quad (9)$$

Our initial approach was to use t_0 as our time of disruption label. However, comparing t_0 to the labeled disruption times $\{t_D\}$ in our test set, we find that most of our predictions are early, with $\Delta t = t_0 - t_D \approx -3.9$ ms (Figure 4a). As seen in Fig 5, the DIII-D team commonly labels shots near the middle of our disruption range. This led us to formulate a third disruption label t_{root} in Eq (10) by finding the root of $f(t)$ where $t_0 \leq t \leq t_f$.

$$f(t_{\text{root}}) = 0, \quad t_{\text{root}} \in [t_0, t_f] \quad (10)$$

If we use the first root t_{root} (Eq (10)) as the predicted disruption time, we see a normalized distribution of $\Delta t = t_{\text{root}} - t_D$ centered around $\Delta t = 0$ in Figure 4b.

Despite t_{root} being closer to the DIII-D labels, we believe that t_0 is a more useful label for disruption mitigation. A disruption forecasting model trained on the start of disruption will be better suited for warning reactor operators before a disruption occurs.

We test our time-of-disruption model on 889 disruptive shots in our test holdout set (Figure 6). Note that we see few late outliers because plasma current drops to zero after a disruption, leaving little room for late predictions. Our errors Δt have a standard deviation of 1 ms, well below

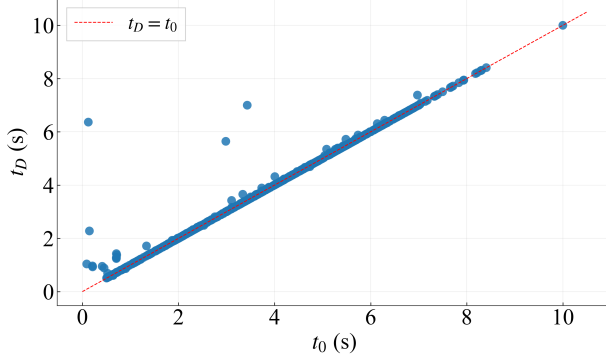


Figure 6: Predicted disruption times in Eq (8) vs real disruption times for 889 test holdout shots.

the 30 ms ITER warning deadline (Figure 4). We therefore expect any time-series forecasting model trained on our generated labels to have a margin of error tolerable within ITER limits.

In the future, we hope to expand our model to produce richer disruption labels. Plasma in a Tokamak can experience a variety of magnetohydrodynamic (MHD) or kinetic instabilities, such as Tearing Modes, Ballooning Modes, Edge Localized Modes, and MHD Kink Modes [30]. Currently, our model can identify a disruption from plasma current alone, but cannot identify the unstable mode behind said disruption. Plasma current alone does not provide sufficient information for making such predictions, so we would likely incorporate multi-modal data streams in future models. For instance, a 2024 paper uses a CNN architecture with 16 input signals, including plasma current and pressure, electron density and temperature, magnetic field strength, and beam power and torque, to produce a likelihood of tearing instability in plasma shots [31]. A sufficiently advanced model could ideally produce a suite of probabilities for entering unstable mode alongside the disruption labels t_0 , t_{root} , t_f that we produce in this paper.

4 Conclusion

We train a multi-layer Convolutional Neural Network to classify shots from the DIII-D Tokamak Reactor as disruptive/non-disruptive using plasma current vs. time data. We use a $F_\beta = 1.5$ performance metric to favor recall over precision and perform hyperparameter tuning over 100 trials; the trained classifier correctly identifies 99% of disruptive shots on our test holdout set, producing less than 9% false positives. We then formulate a convolutional filter to identify the full time-span of a disruption, producing three disruption labels $t_0 < t_{\text{root}} < t_f$. Our time-of-disruption error has a standard deviation of 1 ms, well under the 30 ms ITER warning limit, enabling future time-series forecasting models to train on our generated labels. Our algorithm can generate automated labeling of every DIII-D shot up to the current day, and should generalize well to other Tokamak reactors, providing a

much larger labeled dataset to the scientific community for training plasma disruption forecasting models.

Acknowledgments

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Data and code availability

Our code and model weights are on GitHub [32]. We utilized Cursor [33] and Claude [34] for code development and LaTeX formatting.

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